

Simulation-Based Inference

Exercise 1 Solution

Solution: Logit-Normal Transformation

Let $X \sim \mathcal{N}(0, 1)$ and define the transformation

$$Y = f(X) = \frac{1}{1 + e^{-X}}.$$

(a) Domain and Inverse Transformation

Showing $Y \in (0, 1)$

For any $X \in \mathbb{R}$: The term e^{-X} is always positive. Therefore, the denominator $1 + e^{-X} > 1$, which implies that

$$Y = \frac{1}{1 + e^{-X}} < 1.$$

As $X \rightarrow -\infty$, $e^{-X} \rightarrow \infty$ and hence $Y \rightarrow 0$ (but never reaches 0). As $X \rightarrow \infty$, $e^{-X} \rightarrow 0$ and hence $Y \rightarrow 1$ (but never reaches 1).

Thus, Y takes values strictly in the interval $(0, 1)$.

Deriving the Inverse Transformation

Starting with

$$Y = \frac{1}{1 + e^{-X}},$$

we solve for X :

$$\frac{1}{Y} = 1 + e^{-X}.$$

$$e^{-X} = \frac{1}{Y} - 1 = \frac{1 - Y}{Y}.$$

$$-X = \ln\left(\frac{1 - Y}{Y}\right).$$

Thus, the inverse transformation is:

$$X = \ln\left(\frac{Y}{1 - Y}\right).$$

(b) Deriving the PDF Using the Change-of-Variable Formula

We use the change-of-variable formula. If $Y = f(X)$ with inverse $X = f^{-1}(Y)$, then the density of Y is given by:

$$p_Y(y) = p_X(f^{-1}(y)) \left| \frac{d}{dy} f^{-1}(y) \right|.$$

From part (a), we have:

$$f^{-1}(y) = \ln \left(\frac{y}{1-y} \right).$$

Now, we compute the Derivative (Jacobian). Let

$$g(y) = f^{-1}(y) = \ln \left(\frac{y}{1-y} \right).$$

Differentiate $g(y)$ with respect to y .

Simplify using log rules,

$$g(y) = \ln\left(\frac{y}{1-y}\right) = \ln(y) - \ln(1-y).$$

Differentiate term by term,

$$g'(y) = \frac{1}{y} - \frac{-1}{1-y} = \frac{1}{y} + \frac{1}{1-y}.$$

Combine over a common denominator,

$$g'(y) = \frac{(1-y) + y}{y(1-y)} = \frac{1}{y(1-y)}.$$

The absolute derivative is

$$\left| \frac{d}{dy} f^{-1}(y) \right| = \frac{1}{y(1-y)}.$$

Substitute into the Change-of-Variable Formula. The density of X is the standard normal density:

$$p_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right),$$

where $x = \ln\left(\frac{y}{1-y}\right)$.

Thus, the density of Y is:

$$p_Y(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left[\ln\left(\frac{y}{1-y}\right)\right]^2\right) \cdot \frac{1}{y(1-y)},$$

for $0 < y < 1$.

This is the **logit-normal probability density function (PDF)**.

(a) $Y = \frac{1}{1+e^{-X}}$ takes values in $(0, 1)$ and the inverse transformation is:

$$X = \ln \left(\frac{Y}{1-Y} \right).$$

(b) The PDF of Y using the change-of-variable formula is:

$$p_Y(y) = \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} \left[\ln \left(\frac{y}{1-y} \right) \right]^2 \right) \cdot \frac{1}{y(1-y)}, \quad 0 < y < 1.$$